

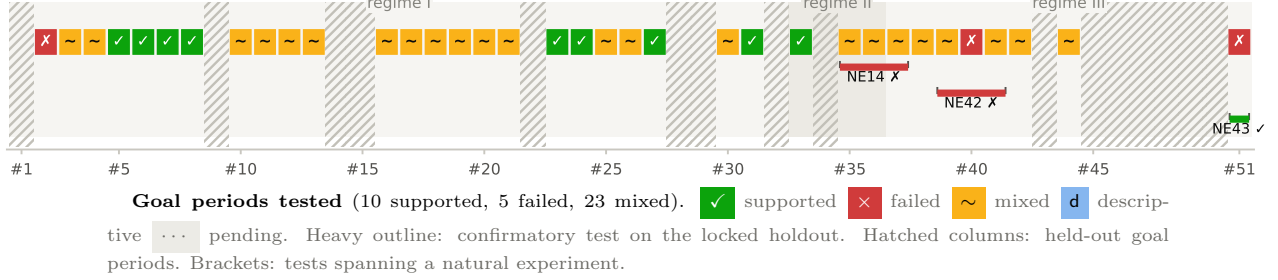
H19: One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

Bottom line. Loop gains do not collapse onto x_{att} ; the round-1 “opposite channels” were the operator’s day edges, and once trimmed all gains move together, rising post hoc with messages per LLM step.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Pooled per-period loop-gain estimates (Hawkes and equal-time methods from H02–H05 plus its own) over 35 non-holdout periods, and asked whether they fall on one curve of x_{att} , attended partners per unit attention. **The pre-registered collapse failed in round 1 and again in round 1b:** no common positive slope, and a regime-only step wins out of sample. Round 1 read this as two channels moving oppositely: activity co-activation rose with x_{att} while talk triggering fell. **Round 1b (corrected minute grid, day-edge-trimmed gains) withdraws that reading:** the activity rise was the operator’s day edges (slope $+0.18 \rightarrow -0.03$ once trimmed; NE14 shows the same at the 2026-03-24 switch). Adjusted, the activity gain agrees with the Hawkes gains, and post hoc all six re-estimated methods rise with messages delivered per LLM step. The unfitted Hawkes-to-Ising mapping (P3) still holds.

PHYSICAL MODEL

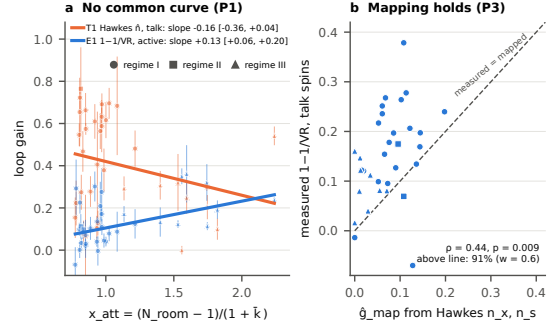
Models 01/02 (Ising) and 09 (Hawkes) in mean field. s_i : agent i ’s 1-min active or talk spin; VR: variance ratio; g_{eq} : equal-time loop gain (H02’s $\beta J_0 q$, H04’s K , H05’s g). n_x, n_s : Hawkes cross-/self-offspring per event; w : share within one $\Delta = 60$ s bin (kernel time τ_k). Dilution: per turn, $\bar{k}$ unread messages from $N_{room} - 1$ partners, response budget b . y_{pm} : method m ’s gain in period p .

$$g_{eq} = 1 - 1/VR, \quad x_{att} = (N_{room} - 1)/(1 + \bar{k})$$

$$VR = \text{Var}(\sum_i s_i) / \sum_i \text{Var} s_i, \quad n_x = b x_{att}$$

$$\hat{g}_{map} = \frac{2wn_x}{1 + 2w(n_x + n_s)}, \quad w = 1 - \frac{\tau_k}{\Delta} (1 - e^{-\Delta/\tau_k})$$

$$y_{pm} = \alpha_m + c_m x_{att,p} + u_p + \varepsilon_{pm}, \quad \text{collapse: } c_m > 0 \forall m$$



Observables. Round 1 (buggy minute grid, untrimmed). (a) The two primary loop gains per period against x_{att} , with their random-effects meta-regression lines: Hawkes talk branching $\hat{n}$ (orange) falls while the equal-time active-spin gain (blue) rises; high x_{att} is essentially regime III (triangles). Round 1b shows the blue rise is day edges (page 2). (b) The unfitted cross-method prediction: the equal-time talk gain against the value mapped from H03’s fast Hawkes kernels ($w = 0.75$); points sit above the identity line.

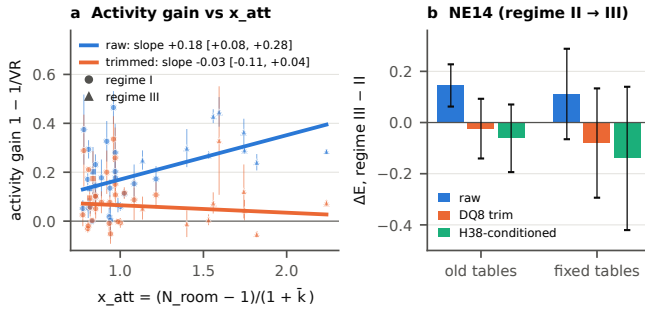
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

A swarm still has no single coupling strength that one configuration knob predicts, so a dashboard “loop gain” would mislead. Round 1b changes the warning, though. What looked like a second, opposite channel was a measurement artifact: the operator starting and stopping every agent at once each day. Once that is trimmed, activity and talk gains move together, and the one candidate knob (chat exposure per model call) lines up all six methods. That knob is post hoc, entangled with the scaffold, and fails inside #51, so it is a hypothesis to test, not a lever yet. The durable practical point is the trimming rule.

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b raw / 1b trimmed)	Verdict
P1 one curve on x_{att}	E1 slope $+0.13 \rightarrow +0.18/-0.03$; T1 -0.16 ; regime-only wins ($-11.5 \rightarrow -8.7/-7.3$ nats)	$\times$
P2 within-family $\rho \geq 0.4$	E: $-0.48 \rightarrow 0.21/0.28$ (0.44 cond.); T: $0.40 \rightarrow 0.37$	H38- $\times \rightarrow$ partial
P3 Hawkes→Ising map (unfitted)	ρ $0.44 \rightarrow 0.37/0.33$; above map 91% $\rightarrow$ 94/89%	$\checkmark$
P4 per-pair $n_x \propto (1 + \alpha_k \bar{k})^{-\alpha}$	$0.20 \rightarrow 0.26$; on N : 1.03	$\times$
P5 no mode effect	CIs include 0	$\checkmark$
Regime III – I, activity gain	$+0.11 \rightarrow +0.17/-0.03$	gone
Native #51 N sweep	per-pair n_c falls far faster than $1/(N-1)$; trimmed g rises with N	$\times$
Native NE42 / NE43 / NE14	talk triggering ≈ 0 in the merged room; edges $\approx \frac{3}{4}$ of g on all NE43 sides; adjusted II→III < 0	$\times \checkmark \times$



Second figure. (a) Activity gain per period against x_{att} , raw (blue) and day-edge-trimmed (orange), with meta-regression lines: the rise disappears once day edges are trimmed. (b) NE14: regime II→III change of the activity gain, raw vs trimmed vs H38-conditioned, on old and fixed tables.

GOAL PERIODS IN DETAIL

- G02 $\times$ 1b: failed corrected data, pre-registered E1; with th...
- G03 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
- G04 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
- G05 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
- G06 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
- G07 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
- G08 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
- G10 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
- G11 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
- G12 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

H04's K and H05's gains (built on the buggy grid) are dropped until their owners re-run them, leaving 6 methods. The trim keeps 77% of regime-III minutes, and two adjustments (DQ8 trim, H38 conditioning) differ in places (NE42). The k_{llm} curve was picked after seeing round 1, its 6/6 agreement is post hoc, and it does not hold within #51. Methods share raw events, so summed ELPD over-counts evidence. The synthetic validation of the P1 rule was not re-run. The frozen confirmatory channel model was fitted on round-1 data and is stale.

ROUND 2

Re-freeze the channel model from round 1b (talk and activity on k_{llm}) and pre-register it; ingest H04/H05 once re-run; test k_{llm} within periods where it varies at fixed roster.

- **Essence.** The swarm has two order parameters, talk and work, that trade off.
- **H19-R1** Talk and work anticorrelate: periods and agents sit in talky or worky phases on a (talk, work) phase diagram.
- **H19-R2** Goal type (shared vs individual) moves the swarm between the two phases.
- **H19-R3** Messages per LLM step is the control parameter; pre-register the post-hoc collapse.

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